Scope Extension Proposal

Adaptive Traffic Signal Control — Minor Project (Batch B1)

Where we are

We are currently building the core software pipeline: a SUMO simulation of a single intersection, a fixed-time baseline for comparison, an adaptive control algorithm chosen from a shortlisted set of candidates, and a mathematically provable fairness guarantee (no approach starves beyond a derived, bounded wait time).

What we'd like to add

A low-cost, beacon-based emergency-vehicle preemption module — layered on top of the same algorithm, not a new subsystem.

- What: an approaching emergency vehicle carries a small RF/Bluetooth beacon; a roadside receiver detects it and signals the controller.
- Why: it plugs directly into the priority-scoring function we are already building — the beacon simply becomes a bounded, temporary override on that approach's score.
- Why it's low-risk: minimal added engineering (a transmitter + receiver + a few lines of override logic) compared to a full sensing or physical-testbed build.

Why this, and why now

- Strengthens our core contribution: proving the fairness bound still holds even under emergency override is genuinely new intellectual content, not just a feature bolt-on.
- Clear real-world value story (emergency response time).
- Best cost-to-novelty ratio of the hardware directions.
- Sets up a natural, larger hardware-in-the-loop and vision-based sensing extension for the project.

Proposed 4-week plan

Week	Focus	Deliverable
1	Finish baseline: fixed-time SUMO sim + metrics pipeline	Working baseline + logged metrics
2	Finalize & implement core adaptive algorithm + fairness bound	Adaptive controller passing fairness stress tests
3	Add beacon-based emergency preemption module	Preemption override integrated into priority scoring
4	Validate, compare vs. baseline, prepare review materials	Full results + report/demo ready

Longer-term roadmap

- Hardware-in-the-loop testbed: a miniature physical intersection (LEDs + microcontroller) driven live by our algorithm, closing the gap between “we simulated it” and “we built it.”
- Edge AI vehicle detection: a camera + lightweight object-detection model (e.g. YOLO on a Raspberry Pi/Jetson) feeding real, noisy vehicle-count data into the same controller — the CCTV-based real-world path our synopsis already points to.
- Pedestrian demand sensing: extending our fairness guarantee to pedestrians as a second demand class, broadening the contribution from vehicle-only to multi-modal fairness-aware control.

Together — a simulated algorithm, proven fair, validated on real hardware with real sensing, and extended with emergency preemption